



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to N.S.K.AGENCIE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

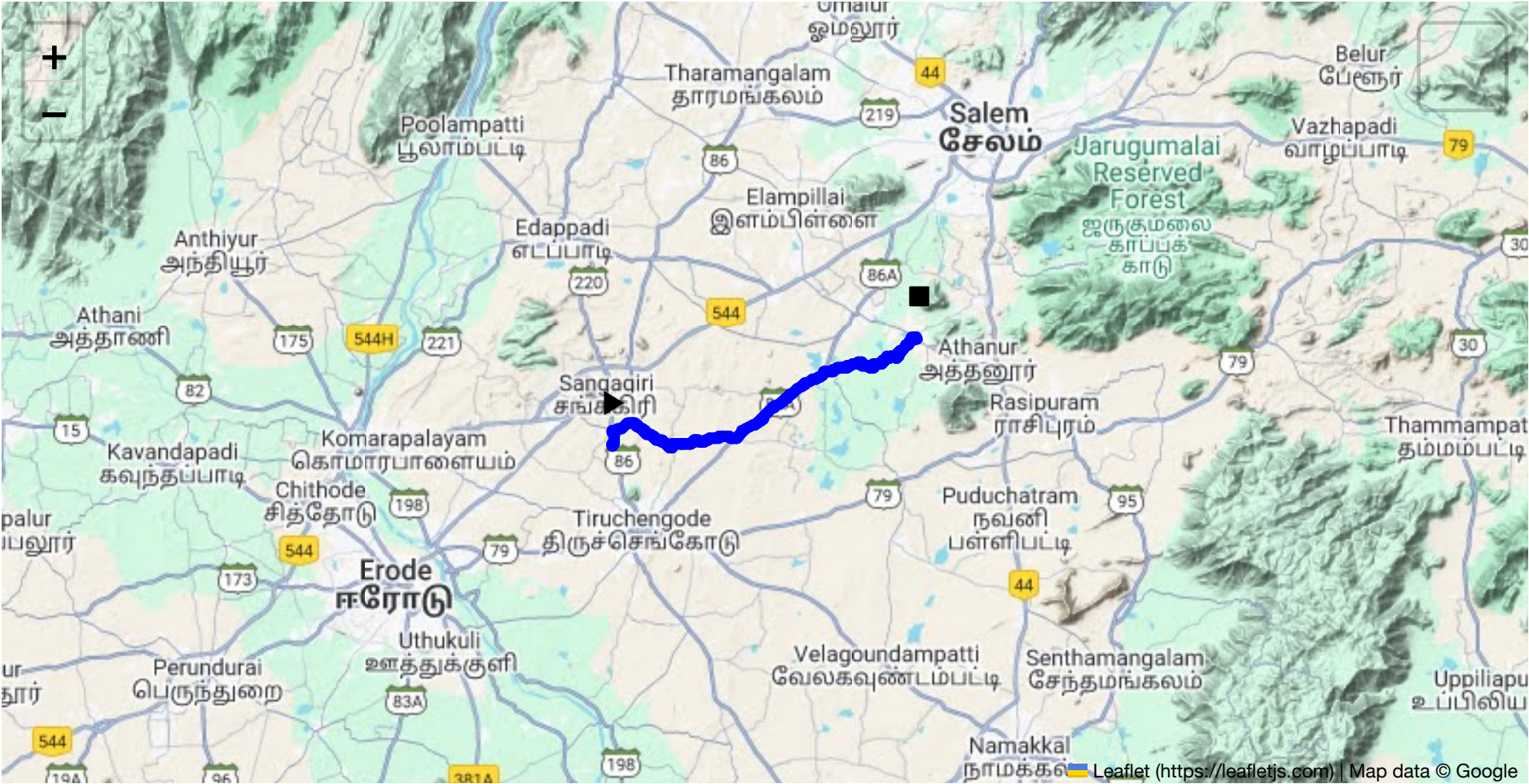
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 30.96 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours

Start: (11.4381, 77.8734)
End: (11.513293, 78.09301)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Vennandur, Tamil Nadu, India

- Overview of the Route Map:** The route from Sangagiri (CVQF+23W) to Vennandur (G37V+953) is approximately 30.96 kilometers. It involves rural and small-town roads typical of Tamil Nadu, with a mix of two-lane highways and smaller roads. Key roads may include state highways and local connecting roads.
- Typical Weather Conditions:** The region experiences a tropical climate with distinct monsoon seasons. High humidity and heat are typical, with heavy rainfall from October to December due to the Northeast monsoon. Weather-related hazards include waterlogging, reduced visibility, and slippery road conditions during monsoons. During summer months, excessive heat can cause road fatigue and affect vehicle performance.
- Traffic Patterns:** Traffic is moderate with peak hours typically in the morning (8 AM - 10 AM) and late afternoon (4 PM - 6 PM). Congestion might be experienced in market towns and near junctions where local traffic intermingles with through traffic. Anticipate slower movement near populated areas such as towns like Edappadi and regions with roadside markets.
- Road Quality and Infrastructure:** Road conditions vary; major roads tend to be well-paved, but expect segments with potholes and uneven surfaces on smaller connecting roads. Infrastructure such as signage and lighting may be less reliable on rural stretches. Narrow bridges and sharp turns in rural areas can pose challenges for larger vehicles.
- Suggestions for Alternative Routes:** In case of emergencies or roadblocks, consider using State Highways that run parallel or provide bypasses around major towns. Locals might recommend shortcuts, but verify their suitability for heavy vehicles.
- Local Regulations for Hazardous Material Transport:** Adherence to Tamil Nadu's regulations for hazardous materials is essential, including permissible hours for transport and required safety measures. Ensure compliance with permits and display of hazardous material placards.

7. **Historical Incidents:** Research reveals few documented incidents involving heavy vehicles; however, minor road mishaps due to road condition and human error are not uncommon. Historical weather-related disruptions have occurred, particularly in monsoon periods.
8. **Environmental Considerations:** Awareness of wildlife and agricultural activities is critical. There are no significant protected environmental zones along the route, but agricultural regions require caution to avoid disruptions to local farming activities.
9. **Communication Coverage:** While most of the route lies within a moderate to good communication coverage zone, rural segments may experience coverage gaps. It's advisable to carry multiple communication devices and ensure emergency contacts are pre-loaded.
10. **Estimated Emergency Response Times:** Emergency response can vary; in populated areas, expect an approximate response time of 20-30 minutes. In rural stretches, this might extend to 45 minutes or more, depending on the location and time of day.
11. **Overall Summary of Risk Assessment:** Moderate risks are present primarily due to weather, road quality variability, and traffic congestion. Major weather events, such as monsoons, pose the greatest risk, while adherence to safety and regulatory practices can mitigate factors related to road quality and traffic. Continuous communication and preparedness for emergency response are crucial to minimizing risk for trucks carrying hazardous materials. Alternative routes should be mapped out in advance and kept accessible for unforeseen circumstances.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45478, 77.88925	30 KM/Hr
5	Turn	High	11.43673, 77.91703	15 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.43967, 77.92345	30 KM/Hr
8	Turn	Medium	11.43979, 77.93009	30 KM/Hr
9	Turn	Medium	11.44030, 77.93066	30 KM/Hr
10	Turn	Medium	11.44067, 77.93203	30 KM/Hr
11	Turn	High	11.44088, 77.93223	15 KM/Hr
12	Turn	Medium	11.44127, 77.93218	30 KM/Hr
13	Turn	High	11.44243, 77.93244	15 KM/Hr
14	Turn	Medium	11.44254, 77.93256	30 KM/Hr
15	Turn	Medium	11.44285, 77.93440	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.44550, 77.95329	30 KM/Hr
17	Turn	Medium	11.44468, 77.95810	30 KM/Hr
18	Turn	Medium	11.44369, 77.96036	30 KM/Hr
19	Turn	Medium	11.44391, 77.96277	30 KM/Hr
20	Turn	High	11.49063, 78.02680	15 KM/Hr
21	Turn	High	11.49135, 78.03252	15 KM/Hr
22	Turn	High	11.49183, 78.03257	15 KM/Hr
23	Turn	Medium	11.49355, 78.03703	30 KM/Hr
24	Turn	Medium	11.49374, 78.04232	30 KM/Hr
25	Turn	Medium	11.49637, 78.04986	30 KM/Hr
26	Turn	High	11.49676, 78.05010	15 KM/Hr
27	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
28	Turn	High	11.49553, 78.05338	15 KM/Hr
29	Turn	Medium	11.49547, 78.05442	30 KM/Hr
30	Turn	High	11.49518, 78.05498	15 KM/Hr
31	Turn	Medium	11.49544, 78.05538	30 KM/Hr
32	Turn	Medium	11.49525, 78.05686	30 KM/Hr
33	Turn	Medium	11.49546, 78.05719	30 KM/Hr
34	Turn	High	11.49548, 78.05904	15 KM/Hr
35	Turn	Medium	11.49477, 78.05905	30 KM/Hr
36	Turn	High	11.49465, 78.05912	15 KM/Hr
37	Turn	Medium	11.49458, 78.05989	30 KM/Hr
38	Turn	High	11.49451, 78.05995	15 KM/Hr
39	Turn	High	11.49309, 78.05982	15 KM/Hr
40	Turn	Medium	11.49303, 78.05997	30 KM/Hr
41	Turn	Medium	11.49306, 78.06011	30 KM/Hr
42	Turn	Medium	11.49362, 78.06091	30 KM/Hr
43	Turn	High	11.49366, 78.06136	15 KM/Hr
44	Turn	Medium	11.49378, 78.06146	30 KM/Hr
45	Turn	Medium	11.49553, 78.06186	30 KM/Hr
46	Turn	Medium	11.49563, 78.06201	30 KM/Hr
47	Turn	Medium	11.49660, 78.06782	30 KM/Hr
48	Turn	Medium	11.49669, 78.06790	30 KM/Hr
49	Turn	High	11.49730, 78.06790	15 KM/Hr
50	Turn	High	11.49738, 78.06877	15 KM/Hr
51	Turn	Medium	11.49906, 78.06934	30 KM/Hr
52	Turn	Medium	11.49914, 78.06944	30 KM/Hr
53	Blind Spot	Blind Spot	11.50124, 78.07191	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
54	Turn	Medium	11.50070, 78.07343	30 KM/Hr
55	Turn	Medium	11.50092, 78.07374	30 KM/Hr
56	Turn	High	11.50082, 78.07865	15 KM/Hr
57	Turn	Medium	11.50374, 78.08048	30 KM/Hr
58	Turn	Medium	11.50424, 78.08040	30 KM/Hr
59	Turn	High	11.50435, 78.08045	15 KM/Hr
60	Turn	High	11.50434, 78.08055	15 KM/Hr
61	Turn	Medium	11.50978, 78.08636	30 KM/Hr
62	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
63	Turn	High	11.51417, 78.09091	15 KM/Hr
64	Turn	Medium	11.51439, 78.09102	30 KM/Hr
65	Blind Spot	Blind Spot	11.51500, 78.09169	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44254, 77.93256



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



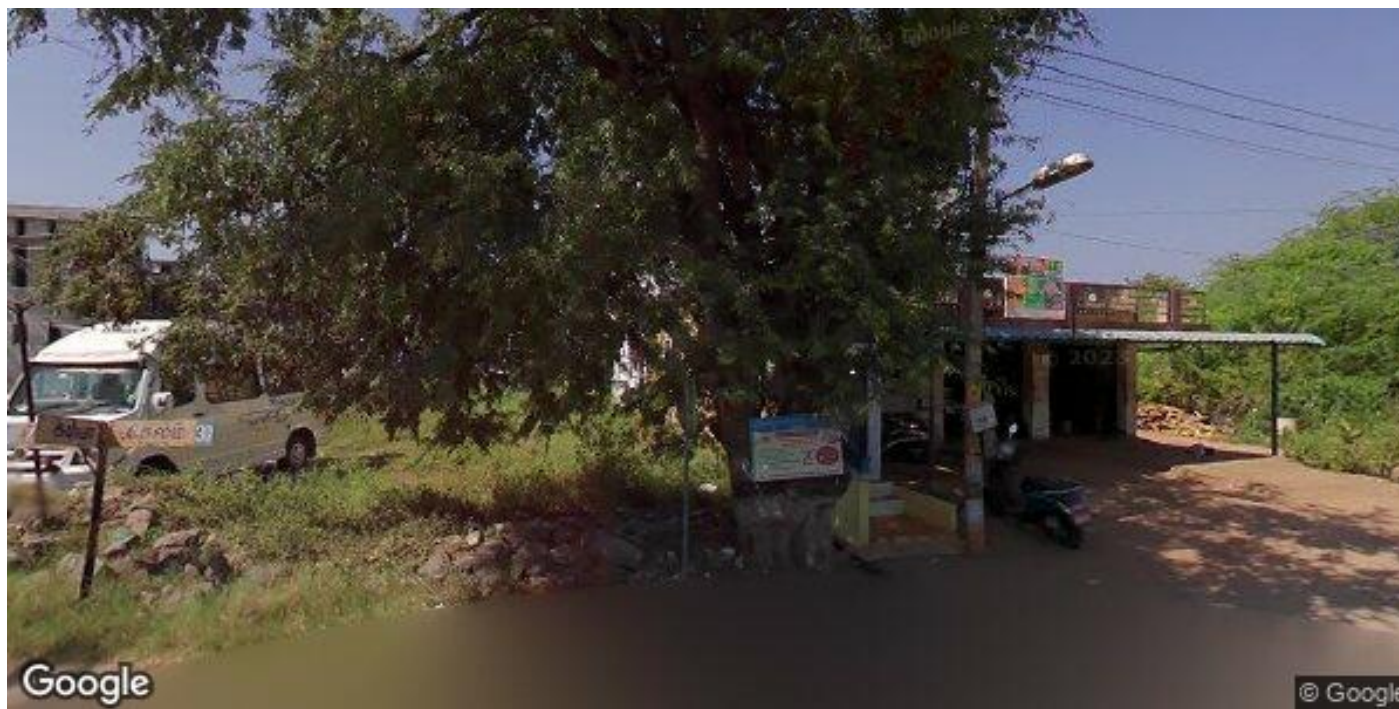
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44468, 77.95810



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44391, 77.96277



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49063, 78.02680



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49135, 78.03252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49183, 78.03257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49355, 78.03703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49374, 78.04232



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49637, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49676, 78.05010



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49553, 78.05338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49518, 78.05498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49544, 78.05538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49525, 78.05686



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49546, 78.05719



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49548, 78.05904



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49477, 78.05905



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 78.05912



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49458, 78.05989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 78.05995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49309, 78.05982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49303, 78.05997



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49306, 78.06011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49362, 78.06091



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49366, 78.06136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49378, 78.06146



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49553, 78.06186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49563, 78.06201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49660, 78.06782



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49669, 78.06790



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49730, 78.06790



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49738, 78.06877



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49906, 78.06934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49914, 78.06944



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.50124, 78.07191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50070, 78.07343



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50092, 78.07374



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50082, 78.07865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50374, 78.08048



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50424, 78.08040



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50435, 78.08045



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50434, 78.08055



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50978, 78.08636



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51417, 78.09091



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51439, 78.09102



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51500, 78.09169
